



Supporting Information

High-Entropy Prussian Blue Analogues and Their Oxide Family as Sulfur Hosts for Lithium-Sulfur Batteries

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Section S1 Experimental Procedures

Materials and instruments

All of the reagents used in this experiment were analytical grade and they used without further purifications. All chemicals, nickel nitrate hexahydrate ($\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$), cobalt nitrate hexahydrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$), copper nitrate trihydrate ($\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$), manganese nitrate tetrahydrate ($\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$), zinc nitrate hexahydrate ($\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$), potassium hexacyanoferrate(III) ($\text{K}_3[\text{Fe}(\text{CN})_6]$), and trisodium citrate dihydrate ($\text{Na}_3\text{C}_6\text{H}_5\text{O}_7 \cdot 2\text{H}_2\text{O}$), were purchased from Shanghai Sinopharm Chemical Reagent Co. Ltd., China. All aqueous solutions were prepared with high-purity de-ionized water (18 MΩ cm).

The phase and crystal structure of the material were characterized by X-ray diffraction (XRD) on a Bruker D8 Advanced X-ray Diffractometer (Cu-Kα radiation: $\lambda = 0.15406$ nm). The morphology of samples was observed by scanning electron microscope (SEM, Zeiss Supra55) under the acceleration voltage of 5.0 kV. Transmission electron microscopy (TEM) investigations were performed by a JEM-2100 instrument. Energy dispersive X-ray spectrometry (EDS) elemental mapping scans were recorded using Tecnai G2 F30 S-TWIN at an acceleration voltage of 300 kV. The chemical states were measured using an Axis Ultra X-ray photoelectron spectroscope (XPS, Thermo Fisher Scientific ESCALAB250Xi) equipped with a standard monochromatic Al-Kα source ($h\nu = 1486.6$ eV). Fourier transform infrared (FTIR) transmission spectra were obtained on a BRUKER-EQUINOX-55 IR spectrophotometer. N_2 adsorption desorption isotherms were obtained on Quantachrome Instruments, Autosorb IQ3. Fe, Co, Ni, Cu, Mn, and Zn K-edge XAS data were performed using SXRMB beam line from Canadian Light Source, Saskatoon, Canada. Their oxides and metal-coordinated phthalocyanine were used to provide the reference points for the relevant elements. The X-ray absorption fine structure spectroscopy (XAFS) data were background subtracted from the overall absorption, normalized, and Fourier transformed by standard procedures within the Athena program. ultraviolet photoelectron spectroscopy (UPS) were performed at the Catalysis and Surface Science Endstation at the BL11U beamline in the National Synchrotron Radiation Laboratory (NSRL) in Hefei, China.

Synthesis of binary PBAs

In a typical procedure, 2 mmol $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and 2.25 mmol $\text{Na}_3\text{C}_6\text{H}_5\text{O}_7 \cdot 2\text{H}_2\text{O}$ were dissolved in 50 mL deionized water to produce clear solution A. Then, 2 mmol $\text{K}_3[\text{Fe}(\text{CN})_6]$ was dissolved in 50 mL deionized water to obtain clear solution B. Afterward, the solution B was directly poured into the solution A under stirring at room temperature. After continuous stirring for 10 minutes, the obtained mixed solution was aged at room temperature for 24 h. The product was collected by centrifugation and washed repeatedly with deionized water and ethanol, respectively. The obtained product was designated CoFe-PBA. Under similar synthetic procedure, NiFe-PBA, CuFe-PBA and MnFe-PBA were fabricated using 2 mmol $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 2 mmol $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ and 2 mmol $\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, respectively. Unlike this, NiFe-PBA was prepared using 200 mL H_2O in solution A and solution B, respectively.

Synthesis of ternary PBAs

The synthesis steps of ternary PBAs are similar with that of binary PBAs, except that two metal salts replace one metal salt. The CoNiFe-PBA, NiCuFe-PBA, CuZnFe-PBA, NiZnFe-PBA, CoMnFe-PBA, NiMnFe-PBA, and CuMnFe-PBA were prepared with 0.5 mmol: 1.5 mmol ratio of Ni/Co, Cu/Ni, Cu/Zn, Zn/Ni, Mn/Co, Ni/Mn, and Mn/Cu. The CoNiFe PBA-2, NiCuFe-PBA-2, CuZnFe-PBA-2, NiZnFe-PBA-2, CoMnFe-PBA-2, NiMnFe-PBA-2, and CuMnFe-PBA-2 were prepared with 1.5 mmol:0.5 mmol ratio of Ni/Co, Cu/Ni, Cu/Zn, Zn/Ni, Mn/Co, Ni/Mn, and Mn/Cu.

Synthesis of quaternary PBAs

The synthesis steps of quaternary PBAs are similar with that of binary PBAs, except that three metal salts replace one metal salt. The CoNiCuFe-PBA, CoNiMnFe-PBA, NiCuMnFe-PBA, and NiCuZnFe-PBA were prepared with 1 mmol:0.5 mmol:0.5 mmol ratio of Co/Ni/Cu, Co/Ni/Mn, Ni/Cu/Mn, and Ni/Cu/Zn.

Synthesis of high-entropy PBA

The synthesis steps of high-entropy PBA are similar with that of binary PBAs, except that five metal salts replace one metal salt. The CoNiCuMnZnFe-PBA were prepared with 0.4 mmol:0.4 mmol:0.4 mmol:0.4 mmol:0.4 mmol ratio of Co/Ni/Cu/Mn/Zn.

Synthesis of cubic metal oxides

The NiFe-PBA, CoFe-PBA, CoNiFe-PBA, NiCuFe-PBA, NiZnFe-PBA, CuZnFe-PBA, CoMnFe-PBA, and CoNiCuMnZnFe-PBA were annealed at 500 °C (2°C min^{-1}) in air for 2 h, for obtaining NiFe_2O_4 , CoFe_2O_4 , $\text{Co}_{1.29}\text{Ni}_{1.71}\text{O}_4/\text{Fe}_2\text{O}_3$, $\text{NiFe}_2\text{O}_4/\text{Cu}_2\text{O}$, $\text{Ni}_{0.8}\text{Zn}_{0.2}\text{O}/\text{Fe}_2\text{O}_3$, $\text{CuO}/\text{ZnO}/\text{Fe}_3\text{O}_4$, $\text{MnCo}_2\text{O}_4/\text{Fe}_2\text{O}_3$, and CoNiCuMnZnFe-oxide, respectively.

Synthesis of cubic metal phosphides

The NiFe PBA and NaH_2PO_2 with a mass ratio of 1:10 were placed in two separate positions in a tube furnace with NaH_2PO_2 at the upstream side. Then, the products were heated up to 350 °C with a heating rate of 2°C min^{-1} and kept at this temperature for 2 h under a N_2 atmosphere. The final products were denoted as $\text{FeP}/\text{Ni}_3\text{P}$, FeP/CoP , $\text{Fe}_3\text{P}/\text{CoP}/\text{Ni}_5\text{P}_2$, and $\text{Fe}_4\text{P}/\text{CoP}_2/\text{Mn}_{1.9}\text{P}$ were prepared using the same method by the pyrolysis of CoFe PBA, CoNiFe-PBA, and CoMnFe-PBA.

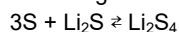
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Synthesis of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S

The sublimated sulfur was successfully introduced into the pores of NiFe-PBA in a quantitative manner through melt diffusion approach at 130 °C. First, 43 mg of NiFe-PBA powder and 100 mg of sublimed sulfur were ground for 30 minutes to obtain a fine mixture. Then the mixture was placed in a 50 mL sealed Teflon autoclave and heated up to 130 °C for 12 h to prepare the final NiFe-PBA-S composites. Moreover, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S were also prepared via the same synthetic procedure.

Adsorption and soaking tests

For the adsorption test, 20 mg NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA were soaked in 3 mL Li_2S_4 solution (10 mmol L^{-1}). The Li_2S_4 solution were prepared in a solvent mixture of 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME) (1:1 in volume) according to the reaction equation:

**Electrochemical Measurements**

Electrode preparation of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S: The slurry was mixed with sample powder, Super P and Polyvinylidene Fluoride (PVDF) in a weight ratio of 7:2:1 in N-methyl-2-pyrrolidone (NMP) as dispersant. Then the slurry was cast on the Al foil (the thickness is 20 nm) and dried overnight at 60 °C under vacuum. The obtained working electrodes were cut to obtain circular electrodes with a diameter of 12 mm. The accurate sulfur mass on each electrode was calculated according to thermogravimetric analysis (TGA) curves under N_2 flow and the elemental analysis data from Elementar, VarioELcube Co. The loading of active sulfur was 0.9~1.1 mg cm^{-2} . The CR 2032-type coin cells were fabricated using the working electrode, lithium foil as the counter and anode electrode, Celgard 2400 as the separator. The electrolyte was used 1.0 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI Sigma-Aldrich (USA), 99.95%) in DOL (Sigma-Aldrich (USA), 99.0%) and DME (Junsei (Japan), 99.0%) (volume ratio, 1:1). 30 μL of electrolyte was used in the fabrication of each Li-S cell in an argon-filled glove box. The value of electrolyte/sulfur (E/S) ratio is $\sim 28 \mu\text{L mg}^{-1}$. The galvanostatic charge/discharge (GCD) tests were estimated in the voltage window of 1.7-2.7 V. The rate capability was also tested by varying the current density from 0.1C to 1C (1C = 1675 mA g^{-1}) on a battery measurement system (CT2001A, Wuhan Land, China) at room temperature. Cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) curves were measured on an electrochemical workstation (CHI660E, Chenhua, Shanghai, China).

In situ UV-vis measurement

The cathode slurry comprised of an active material powder NiFe-PBA-S or HE-PBA-S, Super P, and PVDF with a mass ratio of 7:2:1. Then, the nickel foam was selected as the collector ($1 \times 0.6 \text{ cm}^2$). The sulfur area mass loading on the electrode was 10 mg cm^{-2} . The in situ UV-vis cells were assembled using NiFe-PBA-S or HE-PBA-S electrode as the cathodes and Li metal as the anode, using a custom made in situ cuvette. The in situ cuvette cell was assembled in an Ar filled glove box and sealed in 3 mL of Li-S electrolyte (1.0 mol L^{-1} lithium bis(trifluoromethanesulfonyl)imide (LiTFSI, Sigma-Aldrich, 99.95%) in 1,3-dioxolane (DOL, Sigma-Aldrich, 99.0%) and 1,2-dimethoxyethane (DME, Sigma-Aldrich, 99.0%) (volume ratio, 1:1) with 1 $\omega\%$ LiNO_3). The in situ cuvette cell was galvanostatically discharged at 0.05C. UV-vis spectra were recorded every 15 min from the beginning of the discharge until the end of the discharge for both in situ cuvette batteries. UV-vis absorption spectra (UV-vis, Shimadzu UVmini-1280 spectrophotometer) were used to detect the concentration and elemental chemical states of the LiPS.

In situ XRD measurement

In situ XRD were used to analysis the mechanism of the electrodes during charge/discharge cycles on a X-ray diffractometer (MiniFlex600-C). The sulfur area mass loading on the electrode was 1.3 mg cm^{-2} , and the current collector is Al foil.

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Section S2 Results and discussion

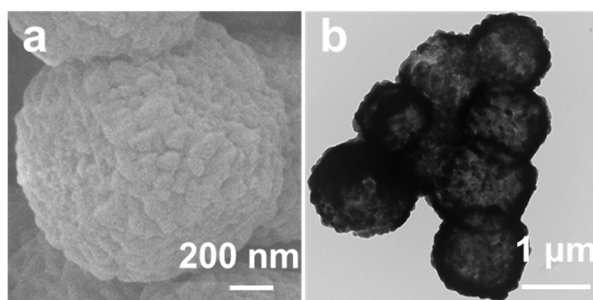


Figure S1. (a) SEM images and (b) TEM images of MnFe-PBA.

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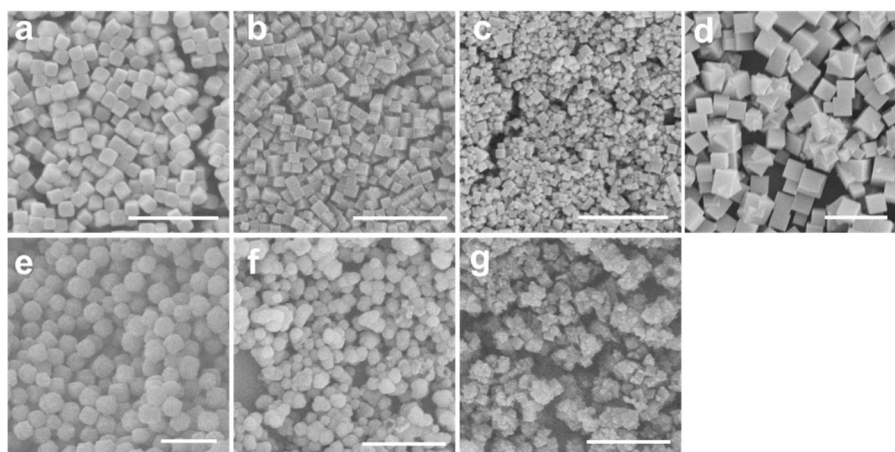


Figure S2. SEM images of (a) CoNiFe-PBA-2, (b) NiCuFe-PBA-2, (c) CuZnFe-PBA-2, (d) NiZnFe-PBA-2, (e) CoMnFe-PBA-2, (f) NiMnFe-PBA-2, and (g) CuMnFe-PBA-2.

The morphology of CoMnFe-PBA-2 with 3:1 molar ratio of Mn/Co exhibits uniform spherical shape. The particle sizes of CoNiFe-PBA-2, NiCuFe-PBA-2, CuZnFe-PBA-2, NiZnFe-PBA-2, CoMnFe-PBA-2, and NiMnFe-PBA-2 are ~170 nm, ~150 nm, ~90 nm, ~300 nm, ~310 nm, ~140 nm. The CuMnFe-PBA-2 did not exhibit a uniform morphology.

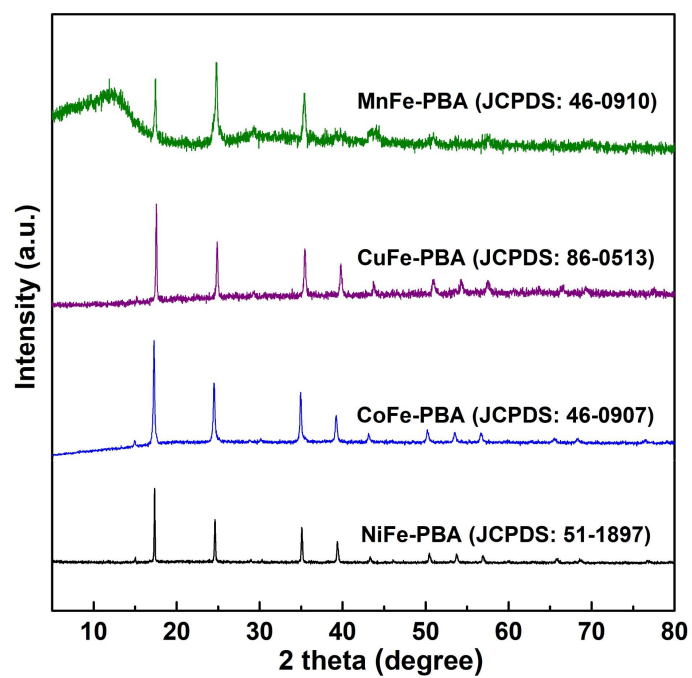
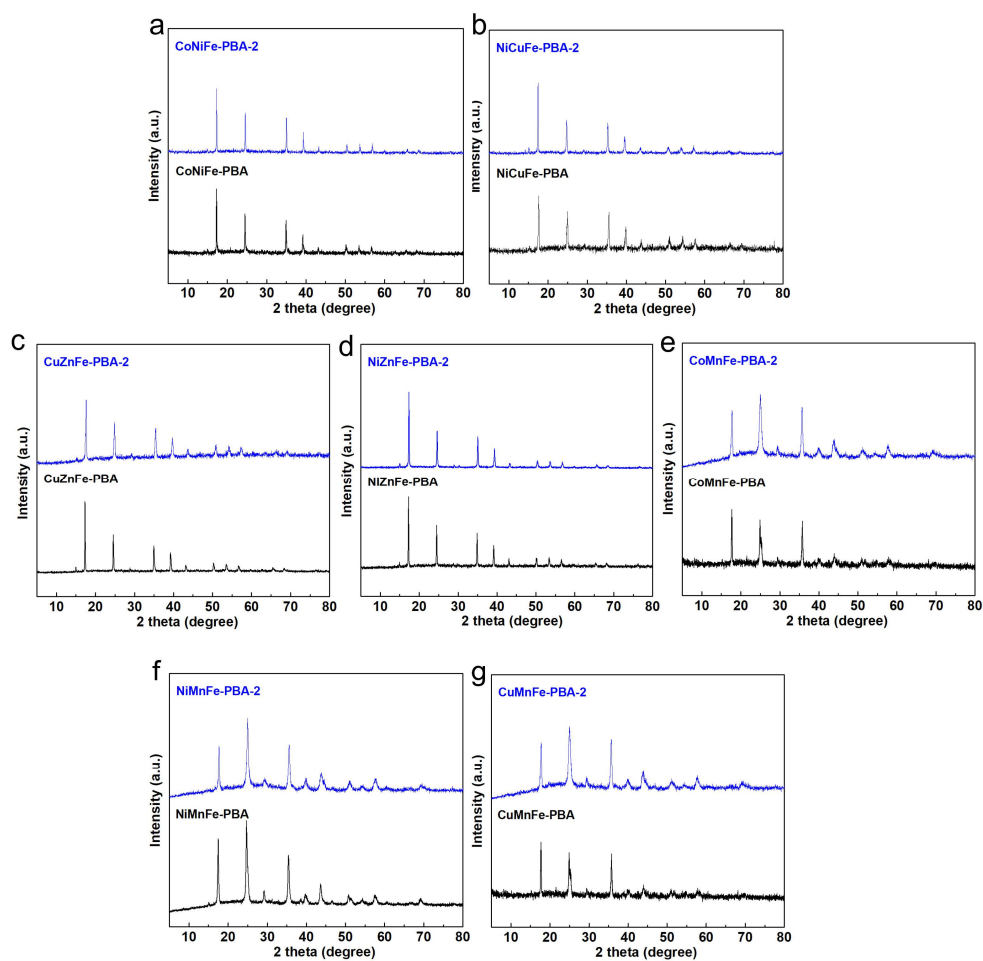


Figure S3. XRD patterns of NiFe-PBA, CoFe-PBA, CuFe-PBA, and MnFe-PBA.

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**Figure S4.** XRD patterns of ternary PBAs.

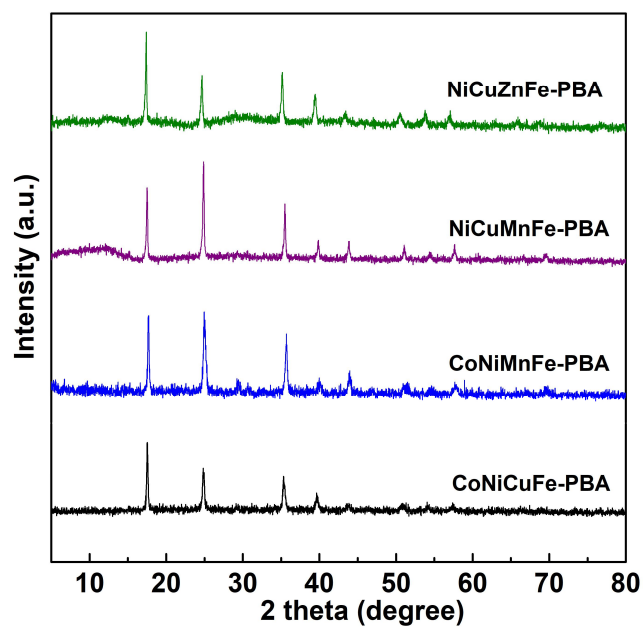


Figure S5. XRD patterns of CoNiCuFe-PBA, CoNiMnFe-PBA, NiCuMnFe-PBA, and NiCuZnFe-PBA.

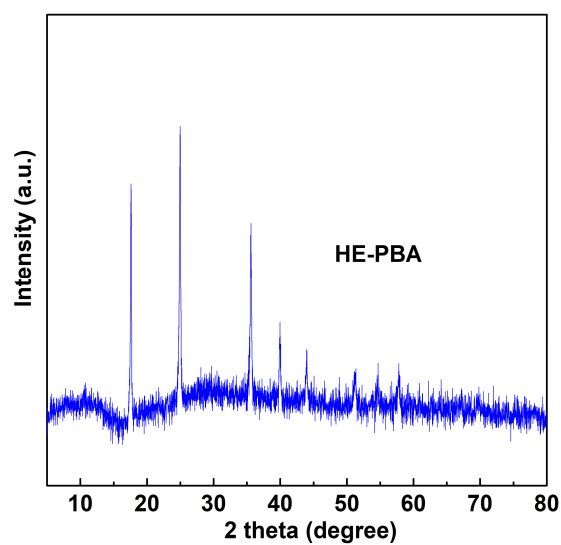


Figure S6. XRD patterns of HE-PBA.

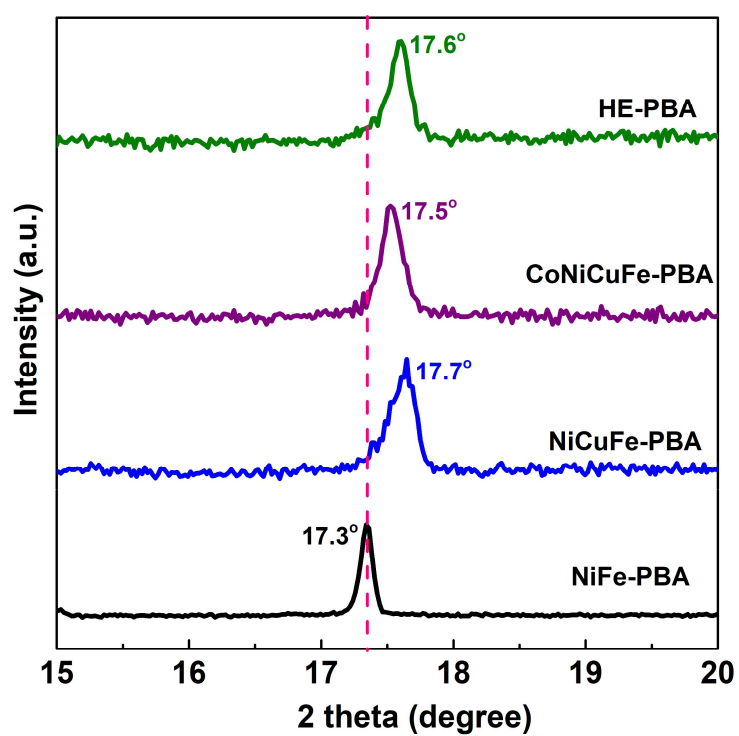


Figure S7. The enlarged view of XRD patterns of NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA.

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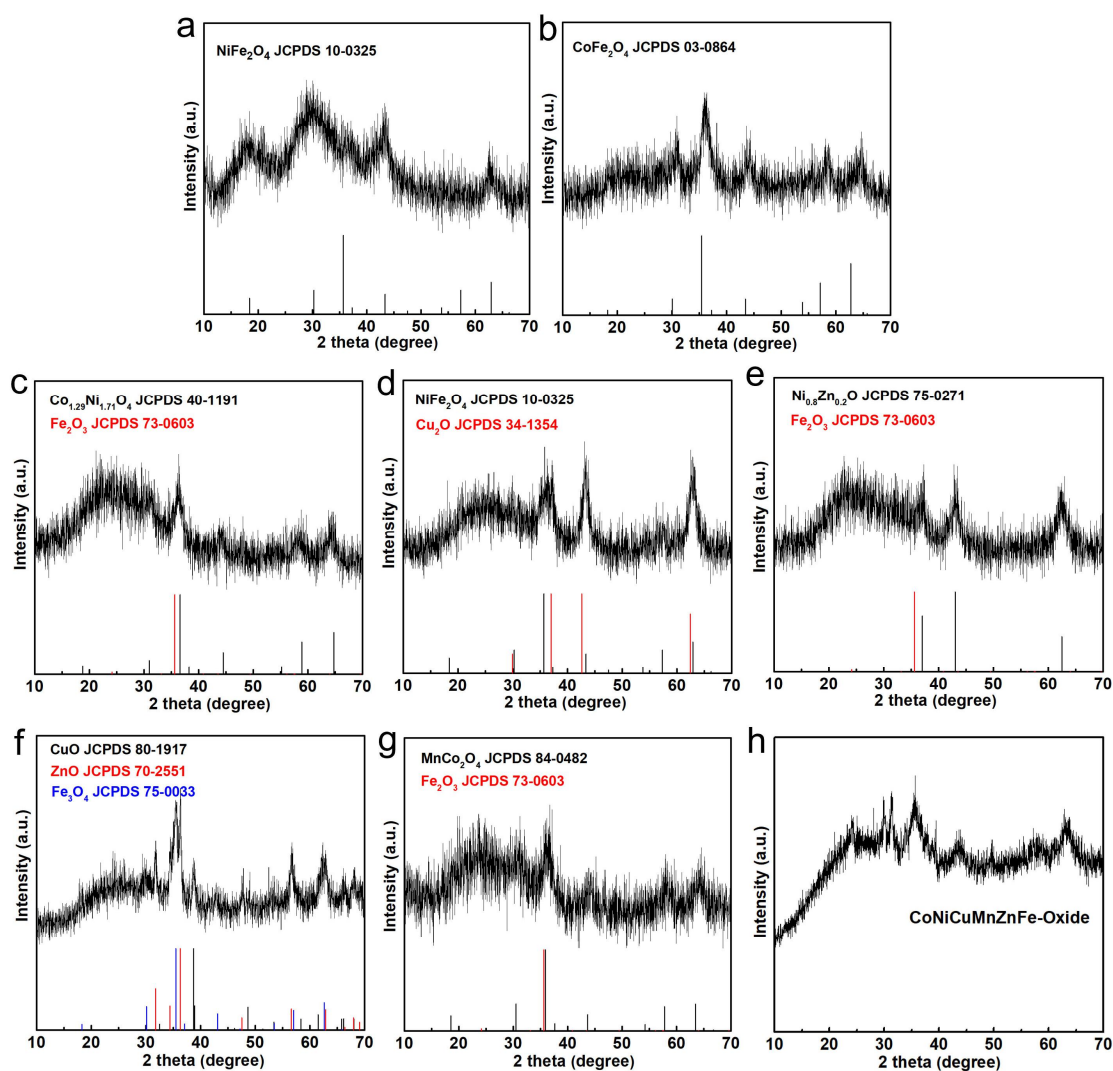


Figure S8. XRD patterns of (a) NiFe_2O_4 , (b) CoFe_2O_4 , (c) $\text{Co}_{1.29}\text{Ni}_{1.71}\text{O}_4/\text{Fe}_2\text{O}_3$, (d) $\text{NiFe}_2\text{O}_4/\text{Cu}_2\text{O}$, (e) $\text{Ni}_{0.8}\text{Zn}_{0.2}\text{O}/\text{Fe}_2\text{O}_3$, (f) $\text{CuO}/\text{ZnO}/\text{Fe}_3\text{O}_4$, (g) $\text{MnCo}_2\text{O}_4/\text{Fe}_2\text{O}_3$, and (h) $\text{CoNiCuMnZnFe-oxide}$.

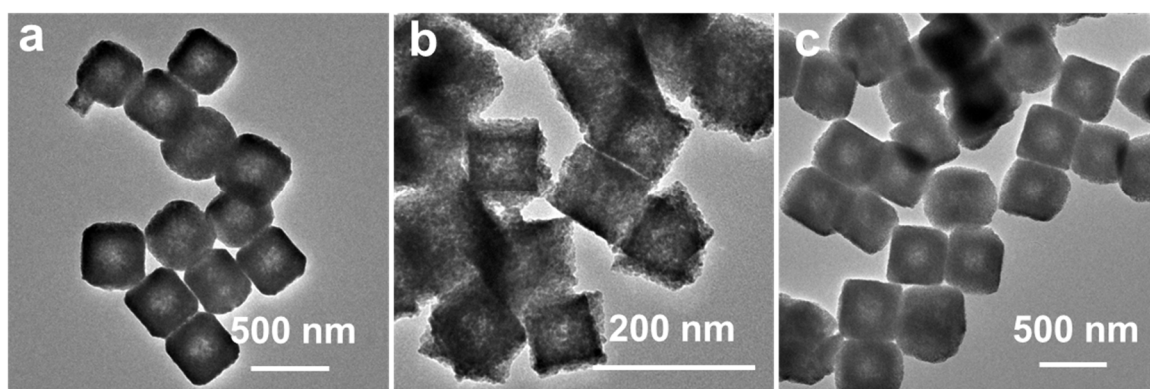


Figure S9. TEM images of (a) $\text{NiFe}_2\text{O}_4/\text{Cu}_2\text{O}$, (b) $\text{Ni}_{0.8}\text{Zn}_{0.2}\text{O}/\text{Fe}_2\text{O}_3$, and (c) $\text{MnCo}_2\text{O}_4/\text{Fe}_2\text{O}_3$.

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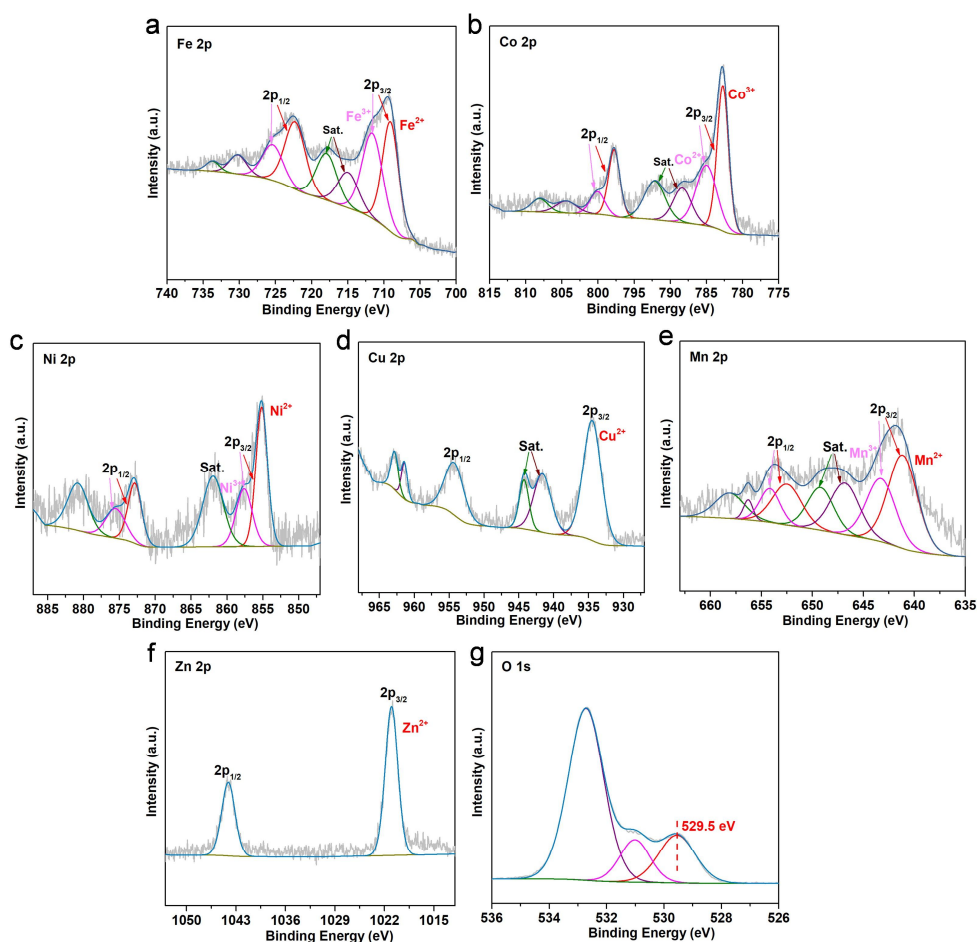


Figure S10. XPS spectra of (a) Fe 2p, (b) Co 2p, (c) Ni 2p, (d) Cu 2p, (e) Mn 2p, (f) Zn 2p, and (g) O 1s.

In the high-resolution Fe 2p spectrum, the peaks located at 725.4 and 722.3 eV were assigned to Fe³⁺ and Fe²⁺ in Fe 2p_{1/2}, while the peak presented at 711.7 and 709.1 eV was contributed to Fe³⁺ and Fe²⁺ in Fe 2p_{3/2} (Figure S10a). In the high-resolution Co 2p spectrum, the binding energies at 785.1 and 800.1 eV correspond to the Co 2p_{3/2} and Co 2p_{1/2} of Co²⁺, respectively, and the binding energies at 782.6 and 797.7 eV correspond to the Co 2p_{3/2} and Co 2p_{1/2} of Co³⁺, respectively (Figure S10b). In the high-resolution Ni 2p spectrum, the binding energies at 855.1 and 872.8 eV can be assigned to the Ni 2p_{3/2} and Ni 2p_{1/2} of Ni²⁺, respectively, and the binding energies at 857.6 and 875.5 eV can be assigned to the Ni 2p_{3/2} and Ni 2p_{1/2} of Ni³⁺ (Figure S10c). In the Cu 2p spectrum, the binding energies at 934.5 and 954.3 eV correspond to the Cu 2p_{3/2} and Cu 2p_{1/2} of Cu²⁺ (Figure S10d). In the high-resolution Mn 2p spectrum, the peaks located at 654.1 and 652.5 eV were assigned to Mn³⁺ and Mn²⁺ in Mn 2p_{1/2}, while the peak presented at 643.3 and 641.2 eV was contributed to Mn³⁺ and Mn²⁺ in Mn 2p_{3/2} (Figure S10e). In the Zn 2p spectrum, the binding energies at 1021.0 and 1044.1 eV correspond to the Zn 2p_{3/2} and Zn 2p_{1/2} of Zn²⁺ (Figure S10f). The O 1s spectra at 529.5 eV corresponds to oxygen atoms bound to metals (Figure S10g).

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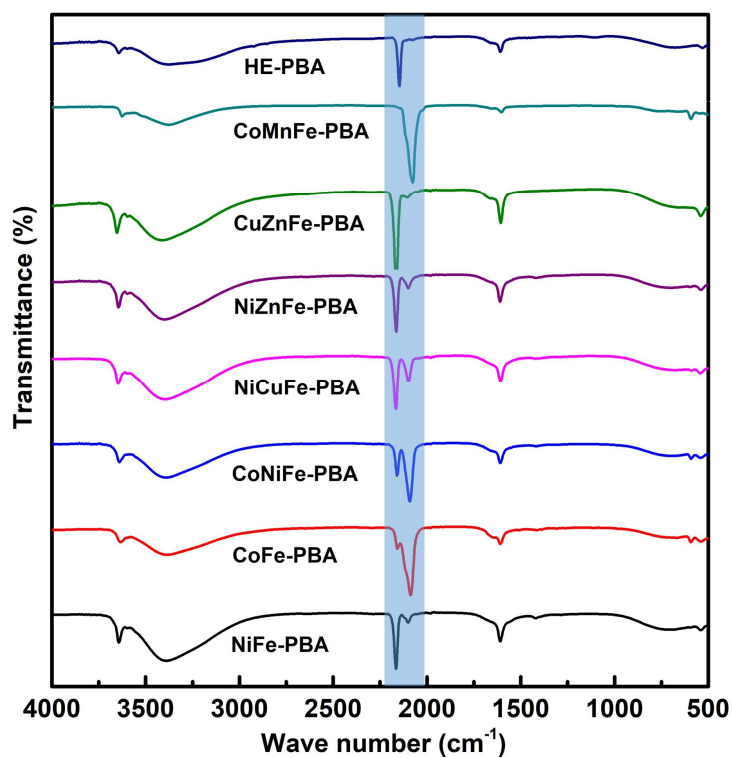


Figure S11. FTIR spectrum of NiFe-PBA, CoFe-PBA, CoNiFe-PBA, NiCuFe-PBA, NiZnFe-PBA, CuZnFe-PBA, CoMnFe-PBA, and HE-PBA.

The peaks located at 3403 and 1607 cm^{-1} result from the -OH stretching vibration and bending vibration of water molecules, respectively. The adsorption bands in the region of 2000–2200 cm^{-1} correspond to the characteristic stretching vibrations of the $\text{C}\equiv\text{N}$ group in the PBA structure.

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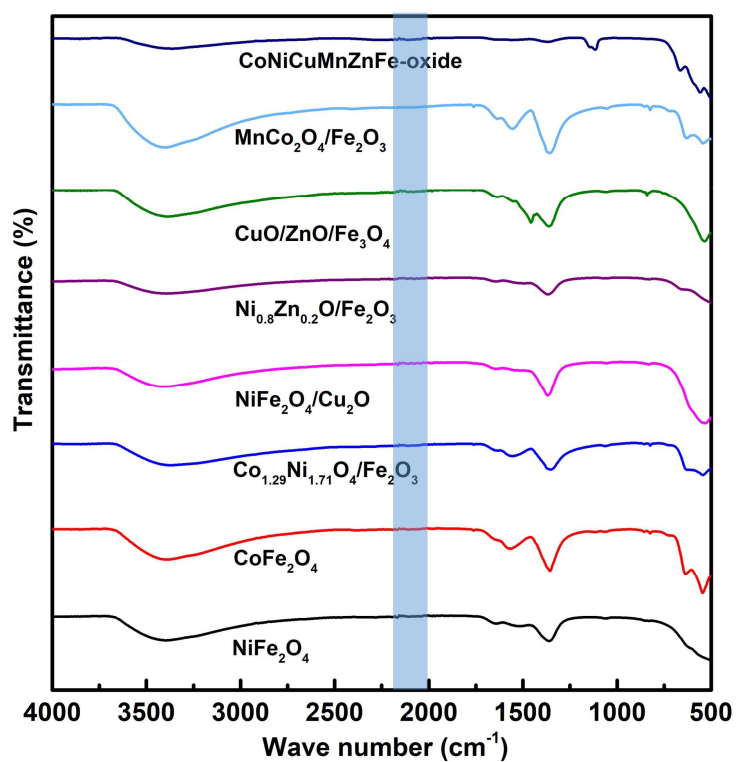


Figure S12. FTIR spectrum of NiFe₂O₄, CoFe₂O₄, Co_{1.29}Ni_{1.71}O₄/Fe₂O₃, NiFe₂O₄/Cu₂O, Ni_{0.8}Zn_{0.2}O/Fe₂O₃, CuO/ZnO/Fe₃O₄, MnCo₂O₄/Fe₂O₃, and CoNiCuMnZnFe-oxide.

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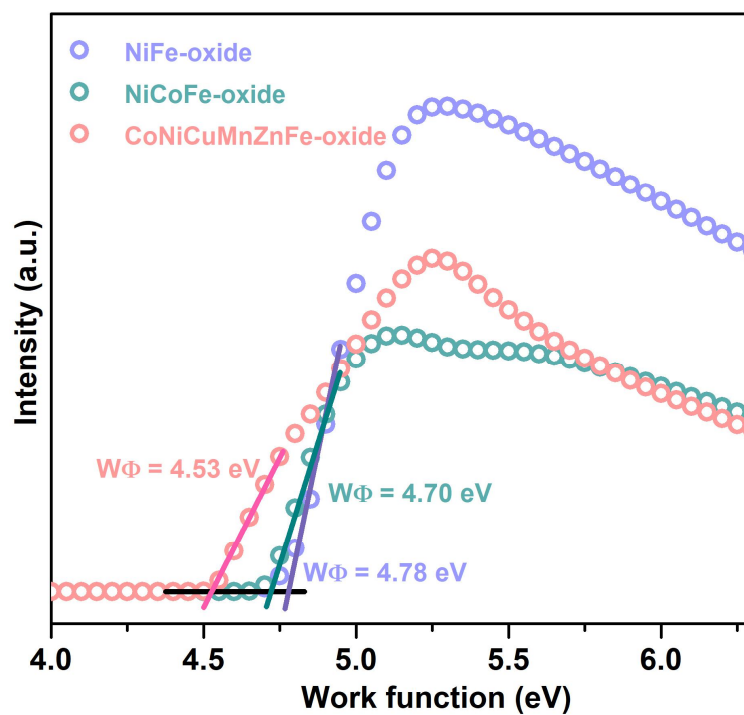


Figure S13. Ultraviolet photoelectron spectroscopy spectra analysis of NiFe-oxide, CoNiFe-oxide, and CoNiCuMnZnFe-oxide.

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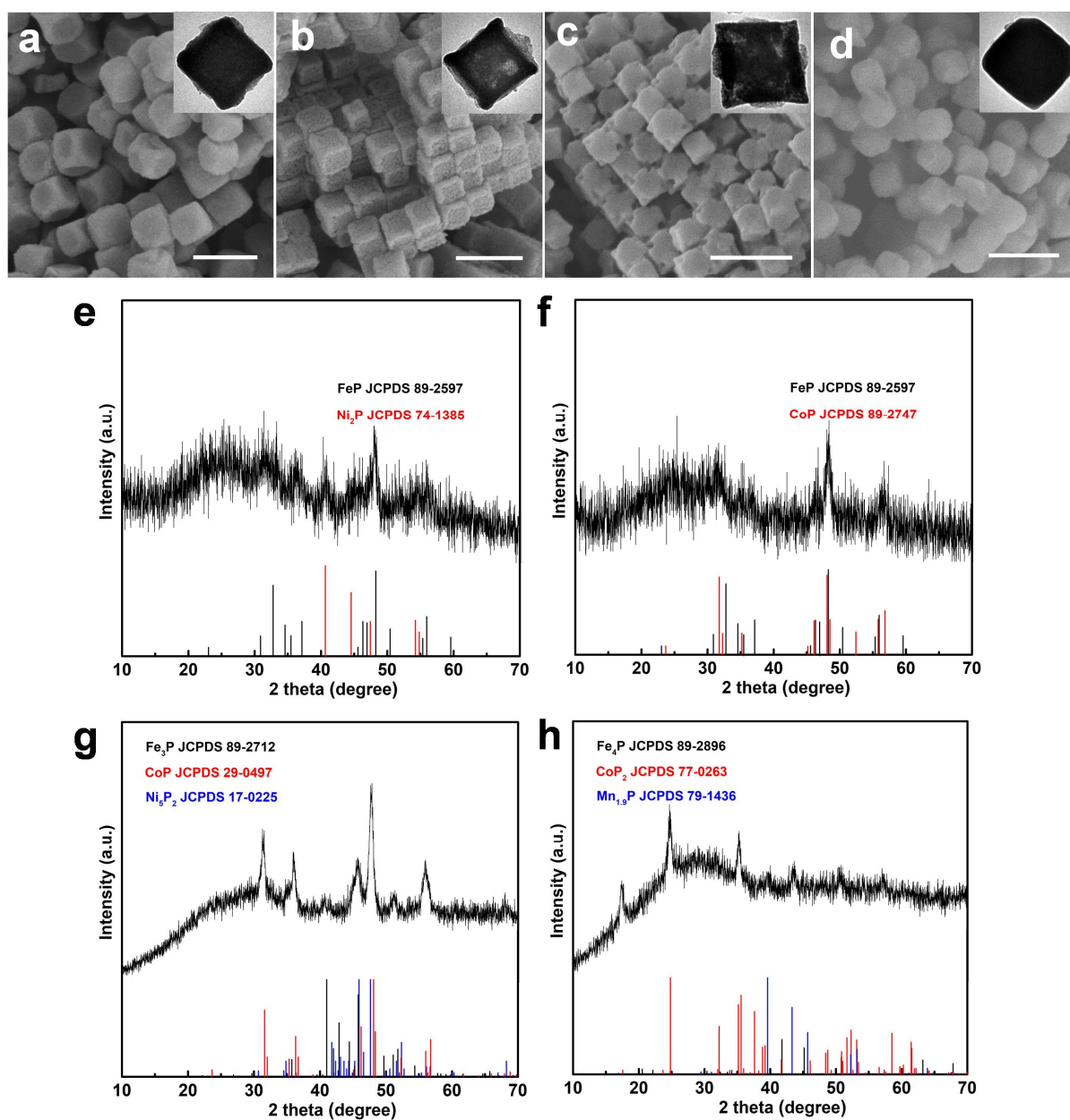


Figure S14. SEM, TEM images of (a) FeP/Ni₂P, (b) FeP/CoP, (c) Fe₃P/CoP/Ni₅P₂, and (d) Fe₄P/CoP₂/Mn_{1.9}P. The scale bars for SEM images indicate 500 nm. XRD patterns of (e) FeP/Ni₂P, (f) FeP/CoP, (g) Fe₃P/CoP/Ni₅P₂, and (h) Fe₄P/CoP₂/Mn_{1.9}P.

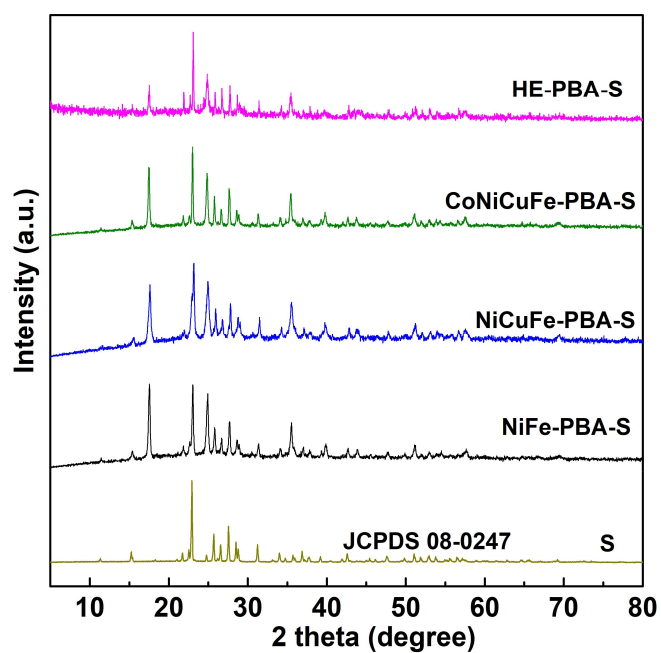


Figure S15. XRD patterns of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S.

After loading the sulfur, the NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S samples still exhibit strong diffraction peaks of the pristine NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA, indicating the stability of their skeleton.

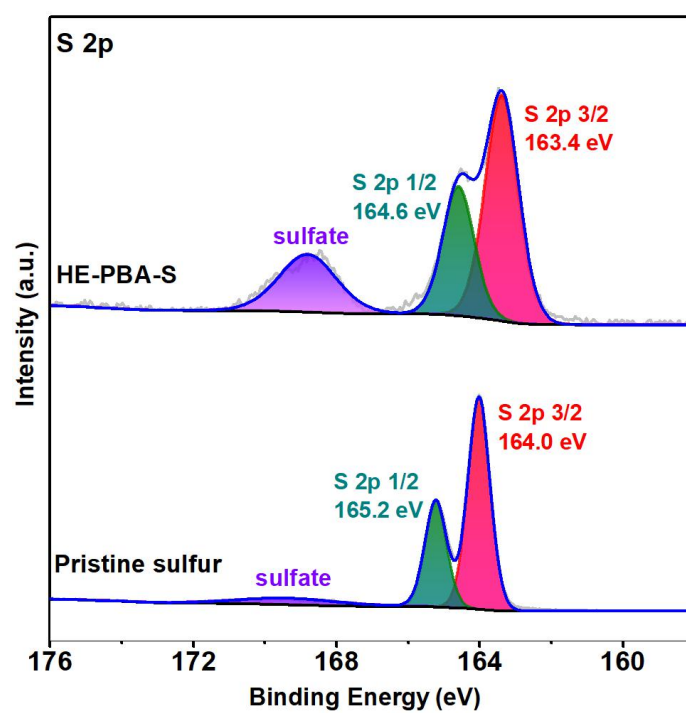


Figure S16. S 2p XPS spectra of pristine sulfur and HE-PBA-S.

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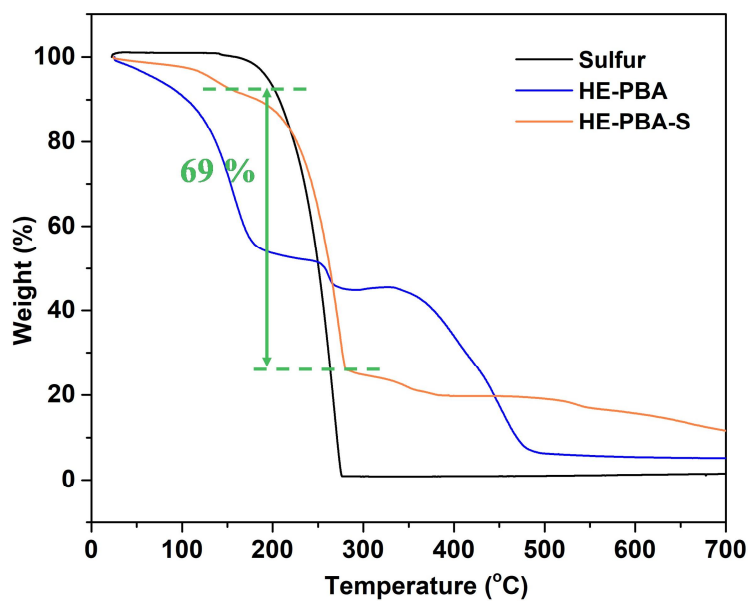


Figure S17. TGA curve of sulfur, HE-PBA and HE-PBA-S.

From the TGA curve, HE-PBA are found to suffer significant quality loss below 170 °C, most likely because of the loss of water. The gradual decrease in weight with temperature increase from 170 to 290 °C results from the removal of coordinating water. A large mass loss observed from 290 to 500 °C can be ascribed to the decomposition of the CN groups.

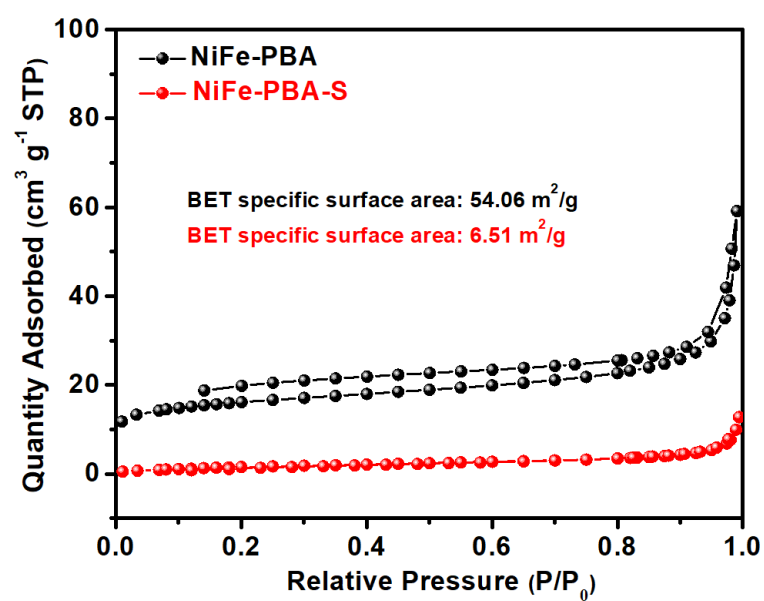


Figure S18. N_2 adsorption-desorption isotherms of NiFe-PBA and NiFe-PBA-S.

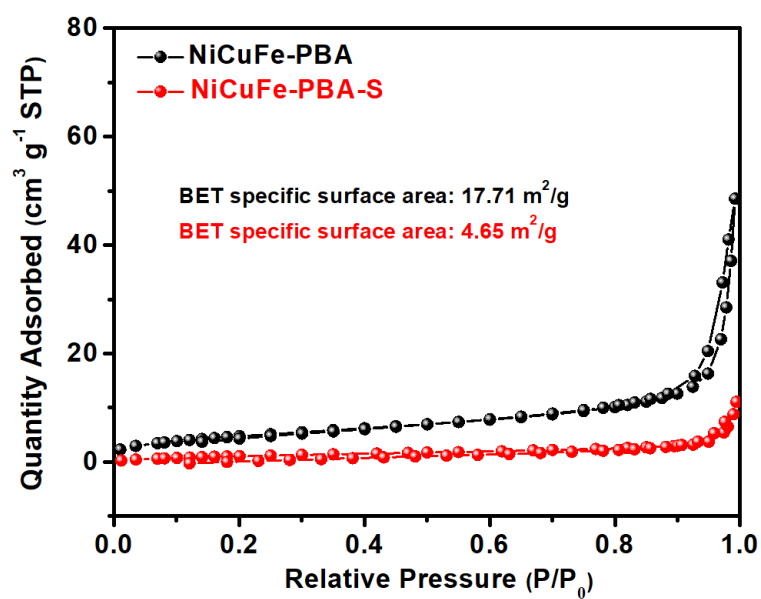


Figure S19. N_2 adsorption-desorption isotherms of NiCuFe-PBA and NiCuFe-PBA-S.

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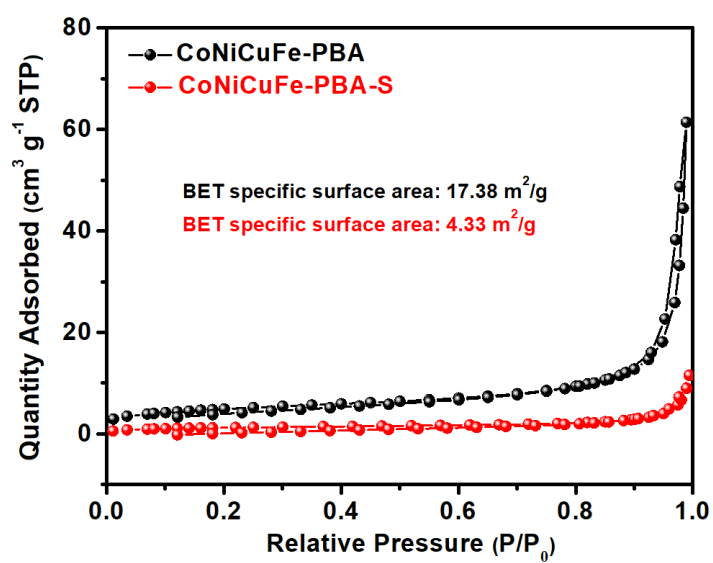


Figure S20. N_2 adsorption-desorption isotherms of CoNiCuFe-PBA and CoNiCuFe-PBA-S.

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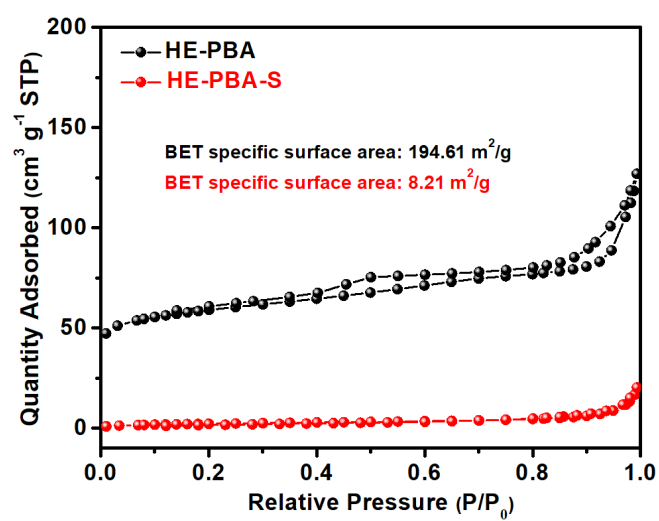


Figure S21. N_2 adsorption-desorption isotherms of HE-PBA and HE-PBA-S.

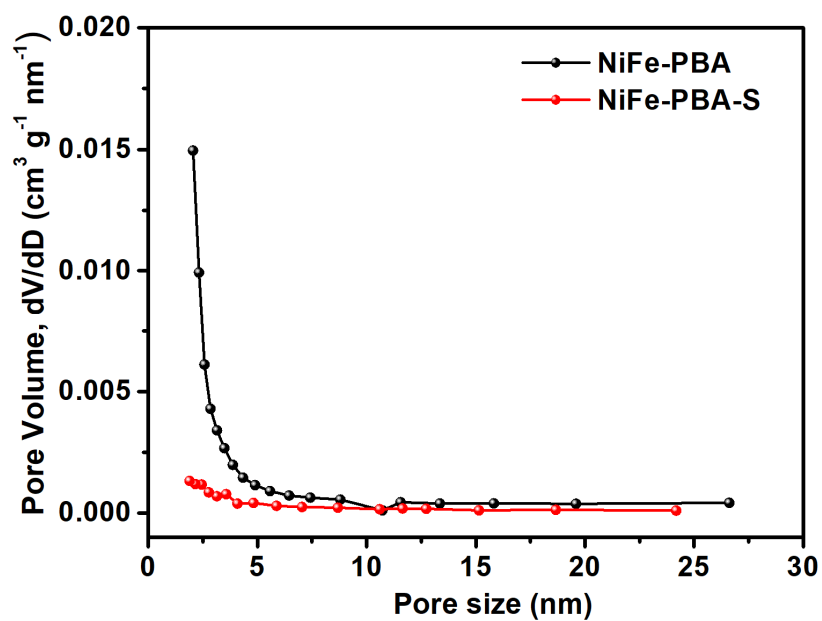


Figure S22. Pore size distribution of NiFe-PBA and NiFe-PBA-S.

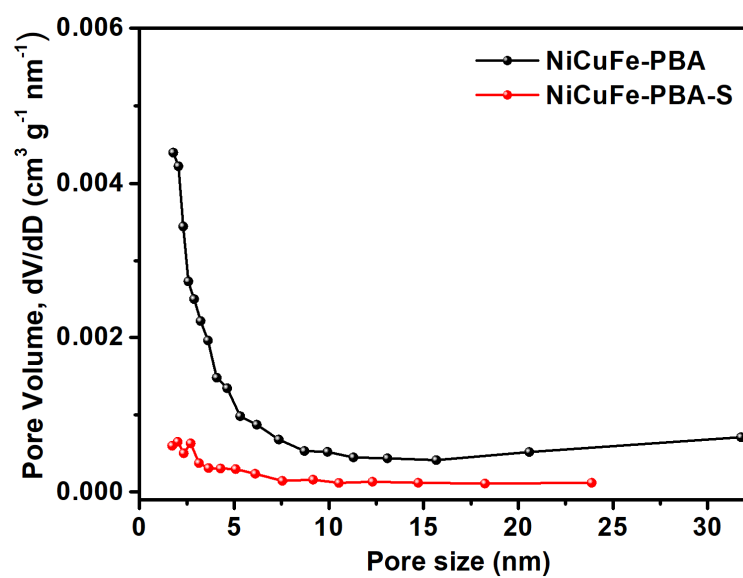


Figure S23. Pore size distribution of NiCuFe-PBA and NiCuFe-PBA-S.

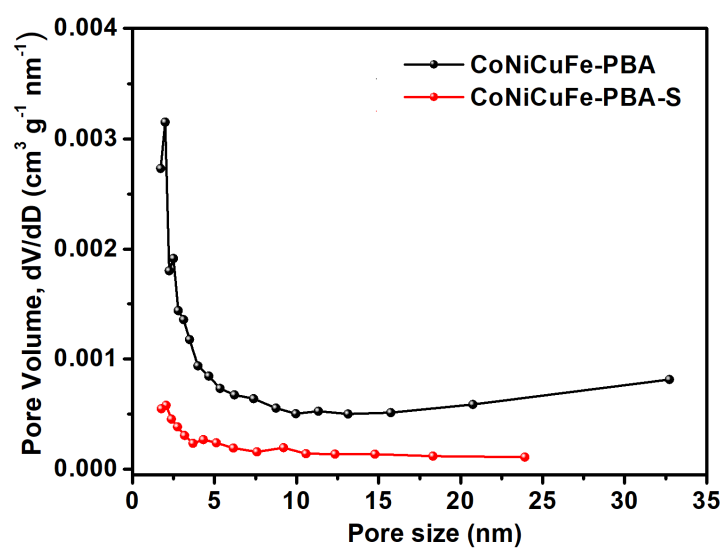


Figure S24. Pore size distribution of CoNiCuFe-PBA and CoNiCuFe-PBA-S.

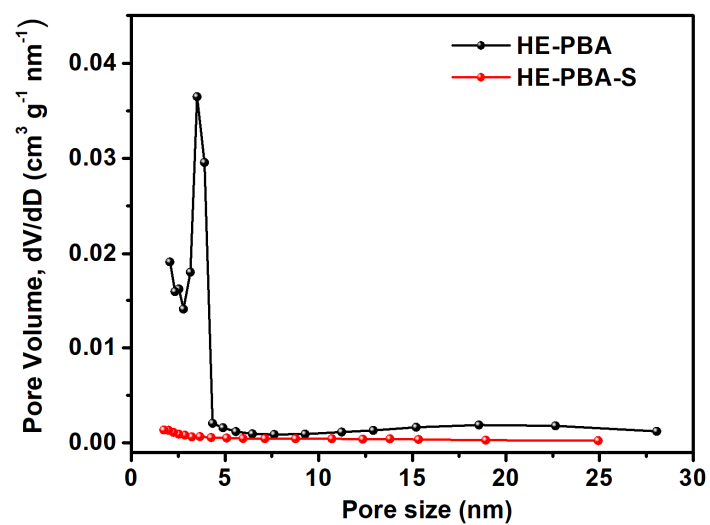


Figure S25. Pore size distribution of HE-PBA and HE-PBA-S.

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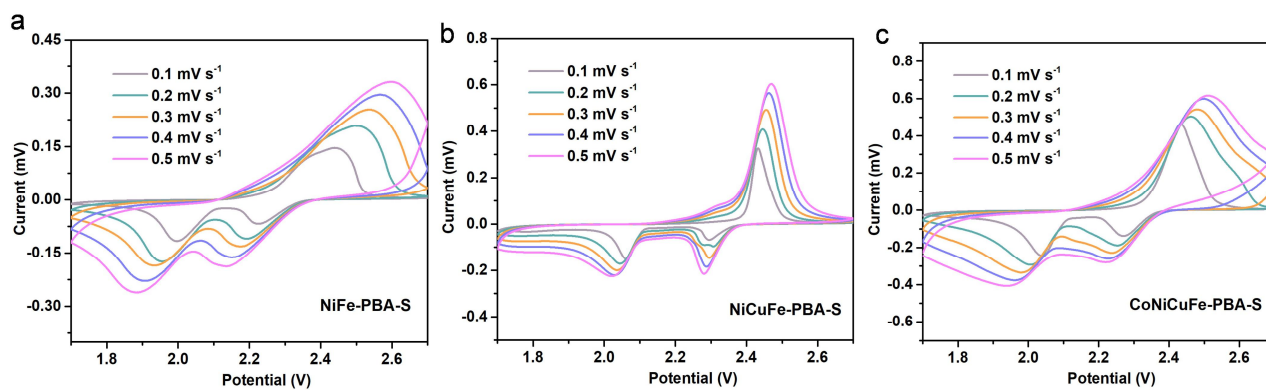


Figure S26. CV curves at different scan rates of (a) NiFe-PBA-S, (b) NiCuFe-PBA-S, and (c) CoNiCuFe-PBA-S electrode.

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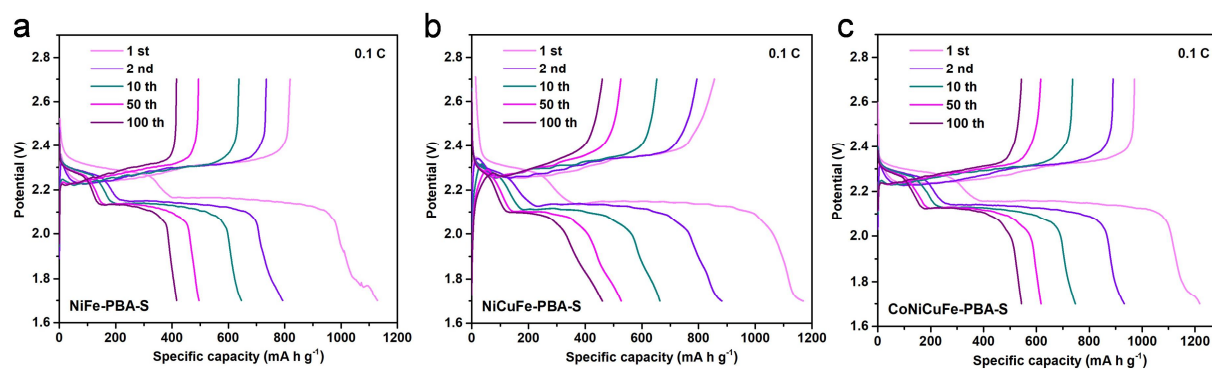


Figure S27. GCD profiles of (a) NiFe-PBA-S, (b) NiCuFe-PBA-S, and (c) CoNiCuFe-PBA-S electrode at 0.1C.

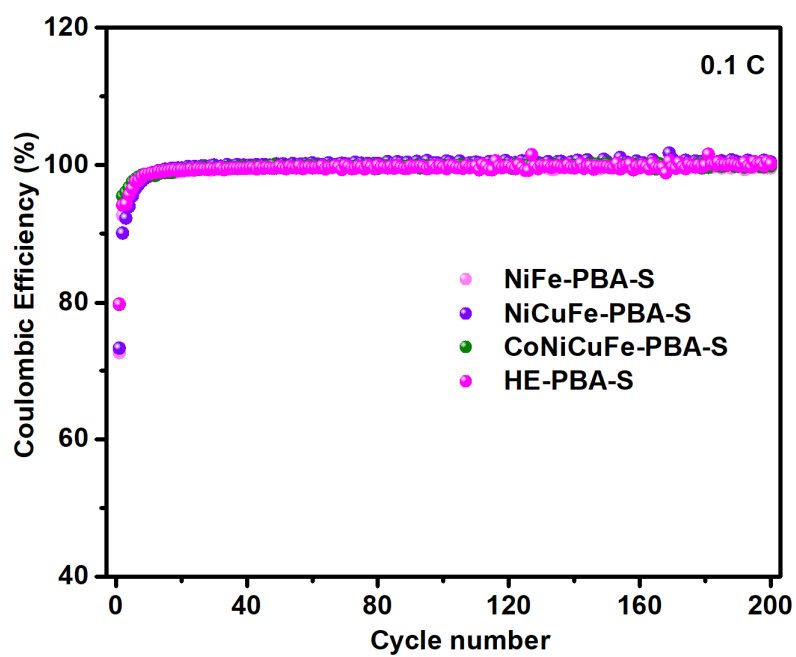


Figure S28. Coulombic efficiency at 0.1C of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S, and HE-PBA-S.

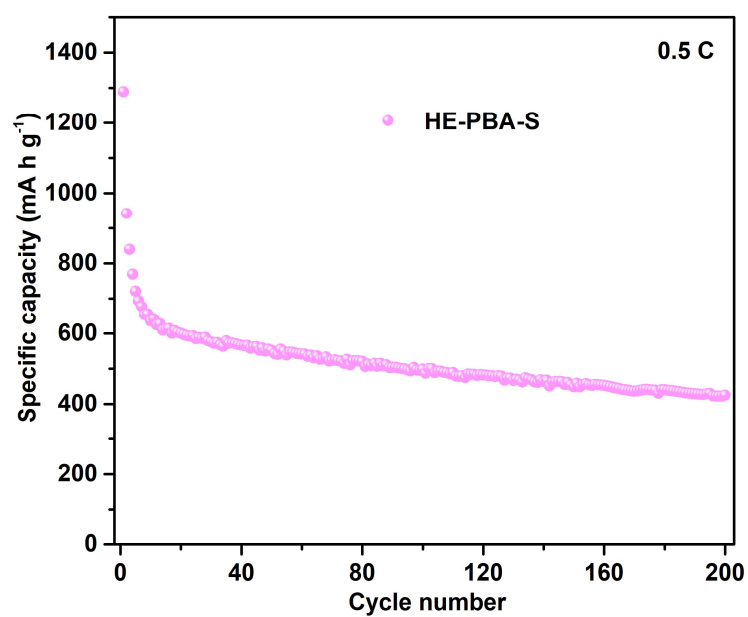


Figure S29. Cycle performance of HE-PBA-S at 0.5C over 200 cycles.

SUPPORTING INFORMATION

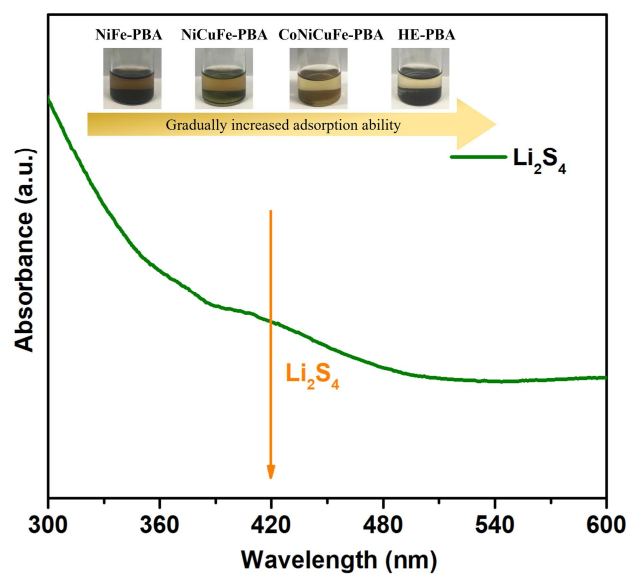


Figure S30. UV-vis spectra of Li_2S_4 solution before adsorption. The absorption peak at approximately 420 nm can be attributed to Li_2S_4 species. The insert shows the visual adsorption test of Li_2S_4 solution on NiFe-PBA, NiCuFe-PBA, CoNiCuFe-PBA, and HE-PBA with the same weight after 4 h.

SUPPORTING INFORMATION

Table S1. ICP-AES Analyses of PBA family.

Samples	Content ratio	Molar ratio (used)	Molar ratio (detected)
NiFe-PBA	Ni:Fe	1:1	0.79:0.90
CoFe-PBA	Co:Fe	1:1	0.64:0.82
CuFe-PBA	Cu:Fe	1:1	0.73:0.89
MnFe-PBA	Mn:Fe	1:1	0.83:0.92
CoNiFe-PBA	Co:Ni:Fe	1.5:0.5:2	1.24:0.33:1.67
NiCuFe-PBA	Ni:Cu:Fe	1.5:0.5:2	0.91:0.34:1.40
CuZnFe-PBA	Zn:Cu:Fe	1.5:0.5:2	0.99:0.41:1.52
NiZnFe-PBA	Ni:Zn:Fe	1.5:0.5:2	1.15:0.39:1.58
CoMnFe-PBA	Co:Mn:Fe	1.5:0.5:2	1.20:0.45:1.73
NiMnFe-PBA	Mn:Ni:Fe	1.5:0.5:2	1.36:0.38:1.80
CuMnFe-PBA	Cu:Mn:Fe	1.5:0.5:2	1.18:0.42:1.66
CoNiCuFe-PBA	Co:Ni:Cu:Fe	1:0.5:0.5:2	0.77:0.39:0.34:1.60
CoNiMnFe-PBA	Co:Ni:Mn:Fe	1:0.5:0.5:2	0.81:0.40:0.48:1.79
NiCuMnFe-PBA	Ni:Cu:Mn:Fe	1:0.5:0.5:2	0.78:0.35:0.40:1.81
NiCuZnFe-PBA	Ni:Cu:Mn:Fe	1:0.5:0.5:2	0.70:0.38:0.44:1.55
CoNiCuMnZnFe-PBA (HE-PBA)	Co:Ni:Cu:Mn:Zn:Fe	0.4:0.4:0.4:0.4:0.4:2	0.32:0.37:0.29:0.38:0.30:1.89
CoNiCuMnZnFe-oxide	Co:Ni:Cu:Mn:Zn:Fe	-	0.29:0.35:0.27:0.35:0.28:1.67

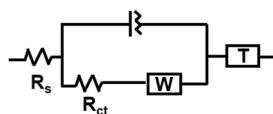
Based on ICP-AES Analyses of PBA family, these observations are in accordance with the molar ratios of the initial feeding materials, further revealing the successful preparation of binary, ternary, quaternary, and high-entropy PBAs, as well as senary metal oxides.

SUPPORTING INFORMATION

Table S2. Elemental analysis data (mass fraction) of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S and HE-PBA-S from Elementar, VarioELcube Co.

Samples	N (%)	C (%)	H (%)	S (%)
NiFe-PBA-S	9.35	8.22	0.72	70.13
NiCuFe-PBA-S	7.95	6.60	0.93	68.74
CoNiCuFe-PBA-S	9.06	7.81	0.85	69.81
HE-PBA-S	7.74	5.92	1.02	70.45

SUPPORTING INFORMATION

Table S3. The equivalent circuit and fitted electrochemical kinetic parameters of NiFe-PBA-S, NiCuFe-PBA-S, CoNiCuFe-PBA-S and HE-PBA-S samples.

Samples	R_s (Ω)	R_{ct} (Ω)
NiFe-PBA-S	3.92	42.46
NiCuFe-PBA-S	4.05	90.17
CoNiCuFe-PBA-S	4.13	24.54
HE-PBA-S	3.38	21.85

SUPPORTING INFORMATION

Table S4. Comparison of the specific capacity for reported sulfur host.

Host Materials	Initial Capacity (mAh g ⁻¹)	Final Capacity (mAh g ⁻¹)	Cycle Number	Discharge Rate (C)	References
S/NH ₂ -MIL-53	568	332	300	0.5	[1]
MIL-100(Cr)/S	1580	~420	60	0.1	[2]
Ni-MOF/S	689	611	100	0.1	[3]
S-in-MIL-53	~750	~500	100	1	[4]
S/FeHCF	1046	~700	50	0.2	[5]
S@Na ₂ Fe[Fe(CN) ₆]	~900	~400	100	0.5	[6]
S@Co-Fe	958	524	100	0.2	[7]
HE-PBA-S	1335.6	570.9	200	0.1	This

SUPPORTING INFORMATION

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